Vishwa R

Site Reliability Engineer

Chennai, India | +91 6382741676 | vishwaravi321@gmail.com | github.com/vishwaravi321 | linkedin.com/in/vishwa-ravichandran

SUMMARY

Site Reliability Engineer with 3+ years operating production Linux systems across multiple clouds (AWS, Hetzner, DigitalOcean), primarily ERPNext/Frappe workloads. Experienced across the reliability stack: incident response and security investigations, infrastructure automation with Terraform and Ansible, CI/CD, monitoring, and backup and disaster recovery. Started out as a developer and still build the Frappe/Python applications I support. Recently building and operating AI-driven automation on top of the platform (LLM agents over fleet data, a driver voice bot), alongside the core reliability work.

EXPERIENCE

Tech Lead, Liqui-Connect Fintech Solutions **Oct 2025 – Present**

Chennai, India

- Own production reliability for ERPNext-based fleet and fintech systems with tight uptime requirements: deployments, rollbacks, tuning, and observability.
- Built PR-based CI/CD pipelines with GitHub Actions that removed the manual server-side release steps and cut deployment time by roughly 80%.
- Traced a recurring CPU spike to the scheduled database backup, then split the application and database tiers and moved backups to a read replica on AWS EC2 with S3 storage, eliminating the recurring CPU saturation the backup caused on the primary.
- Built an AI assistant layer for the fleet stack: an MCP (Model Context Protocol) server that exposes around 40 read-only fleet tools (vehicles, trips, fuel, toll, telemetry, compliance) to Anthropic Claude agents, so operators can query fleet data in plain language.
- Developed and run in production a multilingual driver voice bot that turns phone calls into structured ERPNext records: drivers report issues by phone, calls are transcribed (Sarvam AI) and summarised (Claude), then routed to the right manager, alongside automated pre- and post-trip checklist calls.
- Shipped internal automation on the fleet platform: a WhatsApp bot (Whapi.Cloud) for on-demand vehicle-document retrieval and fuel/expiry alerts, plus BPCL SmartFleet fuel-transaction reconciliation.

Environment: AWS (EC2, S3) • GitHub Actions • Linux • ERPNext/Frappe • Python • MariaDB • Redis • Anthropic Claude (MCP) • Whapi.Cloud • Sarvam AI

Full-Stack Developer, Hyper Ready Technologies **Feb 2025 – Sept 2025**

Coimbatore, India

- Supported and extended production ERPNext for jewellery businesses, improving data correctness and day-to-day stability.
- Wrote custom validations and server-side logic (Python/Frappe) that cut a recurring class of data-entry errors.

Environment: ERPNext/Frappe • Python • MariaDB • Linux

Site Reliability Engineer, Nxweb **Dec 2022 – Dec 2024**

Coimbatore, India

- Sole infrastructure and reliability engineer for 60+ ERPNext production environments running on Hetzner and DigitalOcean.
- Ran ERPNext version upgrades from v12 through v15 with zero data loss, using staging rehearsals and pre-upgrade backups, carried out in planned maintenance windows.
- Automated server provisioning with Terraform and the full Frappe/ERPNext setup with Ansible (roles and Jinja2 templates), which brought new-environment setup down from hours to about 30 minutes and removed the manual, error-prone steps.
- Built the monitoring and alerting stack (Prometheus, Grafana, Alertmanager, Blackbox Exporter) with Telegram alerts, cutting outage detection from 15–20 minutes (customer-reported) to under 5 (machine-detected), and added TLS certificate expiry monitoring.
- Investigated two production security incidents, a compromised server running a cryptominer and another enrolled as a Tor exit node. Rebuilt the affected servers clean and hardened the fleet with key-only SSH, disabled root login, a UFW firewall, non-default ports, and self-hosted secrets management (Passbolt).
- Wrote a custom Frappe app to automate daily off-site backups with 30-day retention after a manual process led to a data-loss incident, and used it to recover a production site during an actual production incident.
- Replaced per-developer cloud servers with a Docker-based local dev platform (frappe_docker, images published to GHCR) and trained the team on it, which reduced cloud spend and support load.
- Migrated 20+ client environments from DigitalOcean to Hetzner with planned maintenance windows and no data loss.

Environment: Terraform • Ansible • Prometheus • Grafana • Alertmanager • Docker • Hetzner • DigitalOcean • Nginx • Linux • Bash

PROJECTS

Kubernetes Infrastructure on Hetzner Cloud

Personal project · Terraform / Infrastructure as Code ·github.com/vishwaravi321/k8s-terraform-hetzner

- Modular Terraform IaC that provisions a Kubernetes-ready cluster on Hetzner Cloud — private network, subnets, SSH keys and control-plane + worker nodes — built as 4 reusable modules wired through outputs, with a single server module reused for both control-plane and workers.
- Horizontal worker scaling via **for_each** over typed object variables (nodes added or removed by editing one tfvars list); provider pinned to an exact version and commits gated by **tflint + pre-commit** hooks (fmt, validate, tflint).

Environment: Terraform • Hetzner Cloud • Kubernetes • Linux

SKILLS

Cloud & Infrastructure: AWS (EC2, S3, VPC, IAM, RDS), Hetzner, DigitalOcean
Infrastructure as Code & Configuration: Terraform, Ansible
Containers: Docker, Docker Compose; Kubernetes (working knowledge)
CI/CD & Version Control: GitHub Actions, Git, Bash scripting
Monitoring & Observability: Prometheus, Grafana, Alertmanager, Blackbox Exporter
Operating Systems & Web: Linux administration, Nginx, systemd
Networking: DNS, TLS/SSL, HTTP/HTTPS, Nginx reverse proxy & load balancing
Databases & Caching: MariaDB / MySQL, Redis
Languages: Python, Bash, JavaScript (Frappe framework)
AI / LLM: Anthropic Claude, Model Context Protocol (MCP), LLM agents & tool integration, speech-to-text (Sarvam AI)
Integrations: WhatsApp (Whapi.Cloud), telephony (Twilio, Exotel), BPCL SmartFleet
Practices: Incident response, root cause analysis, backup & disaster recovery, security hardening, high availability, performance tuning, cost optimization

EDUCATION

B.Sc in Computer Science **2018 – 2021**

Sree Sevugan Annamalai College